

Def. Let $f(x) \in F[x]$. An element $\alpha \in \bar{F}$ s.t. $f(\alpha) = 0$ is called a zero of f of multiplicity v if v is the greatest integer s.t. $(x - \alpha)^v$ divides $f(x)$ in $\bar{F}[x]$.

Thm. Let $f(x) \in F[x]$ be an irreducible polynomial in $F[x]$.

Then, all zeros of $f(x)$ in $\bar{F}$ have the same multiplicity.

Proof. Let α and β be zeros of $f(x)$ in $\bar{F}$. Thus, there exists an isomorphism

$\forall \alpha, \beta: F(\alpha) \rightarrow F(\beta)$ leaving F fixed, and mapping α to β .

Now, we can consider the extend Isomorphism of $\forall \alpha, \beta$

$$\tau: \bar{F} \rightarrow \bar{F}$$

and we can induce a natural isomorphism

$$\tau_x: \bar{F}[x] \rightarrow \bar{F}[x]: \sum a_n x^n \mapsto \sum \tau(a_n) x^n.$$

Since $\forall \alpha, \beta$ leaves F fixed, τ_x also leaves $f(x)$ fixed. But

$$\tau_x((x - \alpha)^v) = (x - \tau_x(\alpha))^v = (x - \beta)^v, \quad \text{for any } v \in \mathbb{N}$$

This implies that multiplicity of $\alpha \leq$ multiplicity of β because

$$\tau_x(f(x)) = f(x) \text{ implies } (x - \alpha)^v \text{ divides } f(x) \text{ then } \tau_x((x - \alpha)^v) = (x - \beta)^v \text{ divides } f(x).$$

And reverse ^{apply} same process to $\forall \alpha, \beta$, we can show the reverse inequality.

Thm. If E/F is a finite extension of F , then $[E:F]$ divides $[E:F]$

Proof. Since E is Algebraic over F , there exist finite elements $\alpha_1, \dots, \alpha_n \in E$ s.t.

$$E = F(\alpha_1, \dots, \alpha_n)$$

For each $1 \leq i \leq n$, let $\text{irr}(\alpha_i, F(\alpha_1, \dots, \alpha_{i-1}))$ have n_i distinct zeros that are all of a common multiplicity v_i , by the above Thm. Now, we can write

$$[F(\alpha_1, \dots, \alpha_i) : F(\alpha_1, \dots, \alpha_{i-1})] = \deg \text{irr}(\alpha_i, F(\alpha_1, \dots, \alpha_{i-1})) = n_i \cdot v_i = [F(\alpha_1, \dots, \alpha_i) : F(\alpha_1, \dots, \alpha_{i-1})] v_i.$$

Meanwhile $[E:F] = \prod_i n_i v_i$ and $[E:F] = \prod_i n_i$

gives the result.

Thm. If K/E is finite and E/F is finite, then

K is separable over F if and only if K is separable over E and E is separable over F .

Proof. Suppose that K is separable over F . Then, $[K:F] = \{K:F\}$.

Moreover, since K/E is finite,

$$[K:F] = [K:E] \cdot [E:F] \quad \text{and} \quad \{K:F\} = \{K:E\} \cdot \{E:F\}$$

Since the index divides the degree,

$$[K:F] / \{K:F\} = \underbrace{[K:E] / \{K:E\}}_{e|N} \cdot \underbrace{[E:F] / \{E:F\}}_{e|N} = 1$$

Thus, these are must be same.

The converse is trivial, because

$$[K:F] = [K:E][E:F] = \{K:E\} \cdot \{E:F\} = \{K:F\}.$$

Corollary. If E/F is finite, then

E is separable over F if and only if for each $\alpha \in E$, α is separable over F .

Proof. Suppose that E is separable over F . Given $\alpha \in E$,

$$F \leq F(\alpha) \leq E$$

Therefore $F(\alpha)$ is separable over F . Conversely, since E/F is finite, for some $\alpha_1, \dots, \alpha_n \in E$,

$$E = F(\alpha_1, \dots, \alpha_n)$$

$$3 \cdot 2 = 6$$

Moreover, since $\text{irr}(\alpha_i, F(\alpha_1, \dots, \alpha_{i-1}))$ divides $\text{irr}(\alpha_i, F)$ in $F(\alpha_1, \dots, \alpha_{i-1})$

and α_i is separable over F , α_i is a zero of $\text{irr}(\alpha_i, F(\alpha_1, \dots, \alpha_{i-1}))$ of multiplicity 1.

This means α_i is separable over $F(\alpha_1, \dots, \alpha_{i-1})$

Summary.

$$\begin{array}{c} F \\ | \\ E \\ | \\ F \end{array} \quad F(x)$$

$$[E:F] = \dim_F E$$

Perfect Field

Lemma. Let $\overline{F}$ be an algebraic closure of a field F .

If a given polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0 \in \overline{F}[x]$

satisfies $(f(x))^m \in F[x]$ and $m \cdot 1 \neq 0$ in F , then $f(x) \in F[x]$.

Proof. Observe that $(f(x))^m = x^{mn} + (m \cdot 1)a_{n-1}x^{mn-1} + \dots + a_0^m$

Def. A field F is called perfect if every finite extension E/F is separable.

Thm. Every field of characteristic zero is perfect.

Proof. Let E/F be a finite extension of F , where $\text{ch}(F) = 0$.

Given $\alpha \in E$, $\text{irr}(\alpha, F)$ factors in $\overline{F}[x]$ into

$$f(x) = \prod_i (x - \alpha_i)^{v_i} \text{ where } \alpha_i \in \overline{F} \text{ are the distinct zeros of } \text{irr}(\alpha, F), \\ \text{with } \alpha_i = \alpha.$$

and since all zeros of irreducible have same multiplicity,

$$= \left(\prod_i (x - \alpha_i) \right)^v \text{ for some } v \in \mathbb{N}.$$

Now, by the Lemma, $\left(\prod_i (x - \alpha_i) \right) \in F[x]$, therefore, $\alpha \in E$ is separable over F .

Since E/F is finite, E is separable over F .

Thm. Every finite field is perfect.

Primitive Element Theorem.

Let E/F be a finite separable extension. Then, there exists $\alpha \in E$ st $E = F(\alpha)$.

Proof. If F is finite field, then E is also finite, therefore there exists $\alpha \in E$ st

$$\langle \alpha \rangle = E \setminus \{0\}$$

Thus, $E = F(\alpha)$. Suppose that F is infinite.

Since E is finite extension, E is generated finitely by $\alpha_1, \dots, \alpha_n \in E$. That is,

$$E = F(\alpha_1, \dots, \alpha_n) \text{ for some } \alpha_1, \dots, \alpha_n \in E.$$

Inductively, we will show that for any $\beta, \gamma \in E$,

$$F(\beta, \gamma) = F(\alpha) \text{ for some } \alpha \in E.$$

Let $\text{irr}(\beta, F)$ have distinct zeros $\beta = \beta_1, \beta_2, \dots, \beta_n$, and

let $\text{irr}(\gamma, F)$ have distinct zeros $\gamma = \gamma_1, \dots, \gamma_m$ in $\bar{F}$.

Since E is separable, all multiplicities are 1. Moreover, F is infinite, therefore

there exists $a \in F$ st $a \neq \frac{\beta_i - \beta}{\gamma - \gamma_j}$ for all $1 \leq i \leq n, 2 \leq j \leq m$.

Let $\alpha = \beta + a \cdot \gamma$. Then, $\alpha = \beta + a \gamma \neq \beta_i + a \gamma_j$, thus

$$\alpha - a \cdot \gamma_j \neq \beta_i \text{ for all } 1 \leq i \leq n, 2 \leq j \leq m.$$

Let $f(x) = \text{irr}(\beta, F)$. Then,

$$h(x) = f(\alpha - ax) \in F(\alpha)[x]$$

Now, $h(\gamma) = f(\beta) = 0$. But, $h(\gamma_j) \neq 0$ for all $2 \leq j \leq m$ because

$$h(\gamma_j) = f(\underbrace{\alpha - a\gamma_j}_{\neq \beta_i})$$

Therefore $h(x)$ and $\text{irr}(\gamma, F)$ have a common factor in $F(\alpha)[x]$, $\text{irr}(\gamma, F(\alpha))$, linear.

Thus $\gamma \in F(\alpha)$ and $\beta = \alpha - a\gamma \in F(\alpha)$.